Interpretable Used-Tesla Asking Prices: Trim Indicators and Model-Specific Registration Effects

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https://github.com/v1852465cs15/nycu-data-mining-hw1-tesla

Abstract

Buyers and small dealers need comparable asking-price estimates when screening used electric vehicles. We study 3,754 European Tesla advertisements using a supplied 3,002/752 training/test partition. Ordinary least squares predicts price in euros from seven structured inputs, six title-derived trim indicators, and three model-by-registration-year interactions. The engineered model reduces dealer-grouped cross-validation MAE from EUR 3,991 to EUR 3,324, and supplied-test MAE from EUR 4,290 to EUR 3,711. A random forest remains better at EUR 3,227. Neither meets our proposed EUR 3,000 test-MAE threshold. Ablations isolate the value of each engineered group, and coefficient and case analyses expose sparse-category uncertainty and extreme-listing failures. Because the test set was inspected during earlier development, these are descriptive benchmark results, not a new blind evaluation.

1 Introduction

Buyers screening used EVs and small dealers setting asking prices need comparable-price references. A Tesla name alone does not identify trim, software or condition. We predict advertised EUR price, not sale price or certified fair value; large gaps should prompt inspection.

Our question is whether listing-time trim flags and model-specific year slopes improve transparent linear estimation beyond raw fields and correlation ranking. We contribute train-only feature analysis, matched ablations with dealer-grouped validation, and uncertainty/failure analysis. The engineered model is linear in its coefficients; we do not claim superiority over the nonlinear baseline.

2 Related Work

The studies below provide context, not head-to-head benchmarks: populations, currencies, targets and evaluation protocols differ.

Study/data	Method and reported result
Pudaruth (2014): 97 Mauritian ads	10-fold kNN; MAE 45,189 rupees (Toyota), 25,741 (Nissan). Year and cylinder volume.
Bukvić et al. (2022): Croatian ads	Linear regression: MAE EUR 1,373.2, RMSE 2,104.7; compared with RF/SVM. Model, year and kilometres.
Madhusudhanan et al. (2024): about 2M B2C records	ProbSAINT, 65 attributes; MAPE 5.6%/5.3% on temporal splits. Includes age, mileage, offer duration.
This study: 3,754 European Tesla ads	OLS with title flags and model-year interactions; test MAE EUR 3,711 on 752 rows.

We use listing-time Tesla fields and first registration, rather than manufacture or offer duration. The comparison retains each study's original units.

3 Task Definition and Data

One row is an AutoScout24 advertisement. The continuous target is its asking price in EUR. The training range is EUR 9,000–175,450; it defines observed coverage, not a clipping rule. Inputs are first-registration year, mileage (km), reported power (kW), equipment count, model, seller country, assistance label and selected title keywords. All are available when screening a listing with its price hidden. Seller identity, URL, collection order and target-bearing free text are not predictors.

The supplied workbook records collection on 5 October 2026 across 188 result pages in eight European countries. It contains 3,754 listings, with the first 3,002 in training and 752 in testing. We retain this original partition for reproducibility. Search-page order is neither randomized sampling nor longitudinal forecasting; a later-market claim would require a temporal dataset. The collector's listing-ID deduplication does not establish unique physical vehicles.

All seven structured inputs are populated in the supplied workbook, so no additional imputation is fitted. Earlier filling and assistance anno-

Variable	Mean	SD	Range
Price (EUR)	35,308	13,937	9,000–175,450
Year	2021.46	2.43	2001–2026
Mileage (km)	88,835	58,372	9–635,000
Power (kW)	300	126	7–1,020
Equipment	55	22	2–119

Table 1: Training-only summary ($n = 3,002$). Inherited fields retain suspect values; range is observed, not a validity bound.

tations are inherited, not independently verified. Numeric fields are standardized using the current training fold; categories are one-hot encoded with unknown categories ignored at prediction. We preserve price extremes, because unusually expensive cars need not be erroneous. One training Model Y has year 2001: the main benchmark retains it to match the shared data, and Section 7 reports an exclusion sensitivity. Reported power and equipment counts are seller fields, not certified specifications; equipment lists can include headings. Exterior colour and range are excluded from the primary model; colour is used only in the additional check E1 (Section 8.1).

4 Exploratory Feature Analysis

Figure 1 uses training rows only. Price is associated with mileage, year and vehicle segment, but each marginal plot mixes trims and countries. Training Pearson correlations with price are -0.403 for mileage, 0.286 for year and 0.557 for power. The year–mileage correlation is -0.594; this motivates examining redundancy rather than interpreting either slope in isolation. We retain power as a predictive seller annotation, while acknowledging its uncertain measurement basis.

Univariate numeric r^2 and categorical correlation ratio η^2 place power, model, mileage, year and country above equipment and assistance. These common-scale scores use raw EUR price, not log-price. We test three explicit hypotheses: **H1**, trim indicators reduce error conditional on structured fields; **H2**, model-specific year interactions improve on a shared slope; **H3**, these two groups outperform raw correlation ranking. Their verdicts come from matched ablations in Section 7 and coefficient interpretation in Section 8, not from these plots alone. Additional checks E1–E4 examine exterior colour (an inherited categorical appearance field), country, model and a squared-power term; these are exploratory checks, not replacements for H1–H3.

5 Method

5.1 Pipeline and feature groups

Figure 2 traces each listing from raw fields to a EUR prediction. **Set A** contains the seven structured fields. **Set B** retains the top k fields ranked inside each training fold: squared Pearson correlation for numeric fields and between-category/total target sum of squares for nominal fields. Searching $k = 1, \ldots, 7$, the smallest value within one standard error ($s/\sqrt{5}$) of the lowest CV MAE is $k = 5$, favoring a smaller representation when gains are within fold uncertainty.

Set C, our final OLS representation, keeps all seven raw fields and adds two groups. Six case-insensitive title flags mark Performance/Plaid, Long Range, AWD/Dual Motor, entry trim (RWD/Standard Range), new generation (Highland/Juniper/Plaid/refresh) and HW4. These capture within-model differences that power alone may miss (H1). Flags are mentions, not verified configurations: Plaid activates two flags, and Standard Range does not certify rear-wheel drive. The reproducibility package preserves exact expressions.

Three interactions allow Model S, X and Y to have different registration slopes:

$$u_m = \mathbf{1}(\text{model} = m)(\text{year} - \overline{\text{year}}_{\text{fit}}). \quad (1)$$

They address H2. The fit-partition mean prevents validation-dependent centering. Model 3 shares the base slope with rare models; these latter classes cannot support reliable separate slopes. This is a registration-year association, not observed longitudinal depreciation.

5.2 Linear model and alternatives

OLS minimizes $\sum_i (y_i - b - \mathbf{z}_i^T \boldsymbol{\beta})^2$. It offers direct EUR effects and a small fixed representation, while interactions allow non-additive raw-input behavior. It has no regularization parameter. The frozen model uses no target logarithm or post-hoc clipping; the separate power-squared check does not change it. Full one-hot fitting uses a least-squares solution; inference uses an equivalent full-rank reference coding. The retained representation has 29 coefficients including the intercept.

Ridge is a plausible alternative for correlated features. A training-only dealer-grouped sensitivity searches $\alpha \in \{.01, .1, 1, 10, 100, 1000\}$ on the same Set C inputs; the lowest MAE is EUR 3,324

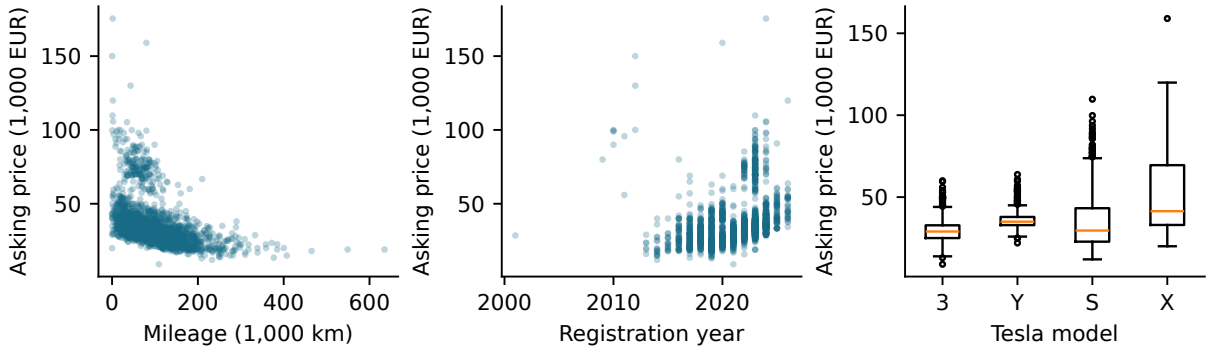


Figure 1: Training-only price against mileage, first-registration year and Tesla model ($n = 3,002$). Units are shown on every axis. Scatterplots retain extremes; boxes show quartiles, with 1.5-IQR whiskers. Association is not a causal depreciation estimate.

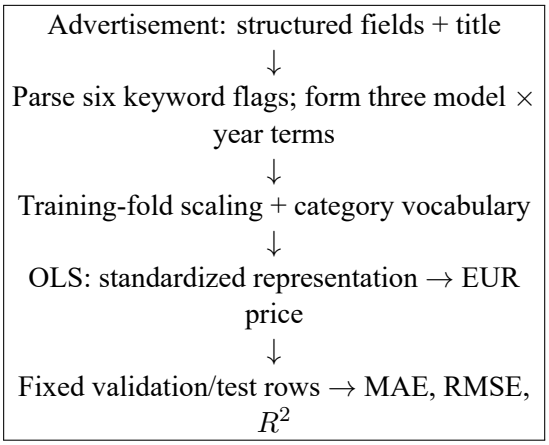


Figure 2: Prediction pipeline. Every learned transformation uses only its fit partition. The target enters regression fitting and evaluation, never feature construction.

at $\alpha = 0.01$, compared with OLS EUR 3,324. We retain the already specified OLS benchmark and report this alternative without choosing a new model from test outcomes. Coefficient uncertainty is conditional on this representation.

6 Experimental Setup

6.1 Validation and provenance

We use identical five-fold shuffled KFold partitions (seed 42) across methods, plus five-fold GroupKFold by seller/dealer as the structure-aware check. Five folds retain about 80% for fitting while permitting a spread estimate at modest cost. There are 1,749 training dealer groups; each missing seller receives its own group. Scaling, encoding, centering and top-k ranking are fitted within folds. We report mean ± sample SD. Feature-set/k exploration used these training folds, so CV is development evidence, not a nested estimate of the entire selection process.

The supplied test set was already inspected in earlier team analyses. We freeze this integration's model definitions before recomputation and retain all 752 rows, but do not claim a new one-use blind holdout. Dealer grouping is not a vehicle-identity guarantee; overlap and sensitivity checks are reported separately. A newly collected locked test set remains necessary for confirmatory deployment claims.

6.2 Metrics and baselines

MAE measures typical absolute EUR error; RMSE penalizes large misses, and R^2 describes explained variation. We use MAE $\leq$ EUR 3,000 as an assumed benchmark tolerance. No stakeholder cost data were collected: this is neither an inspection fee nor a validated buyer/dealer cost. Pass/fail and within-threshold shares are descriptive, not evidence of economic benefit or an individual guarantee.

Baselines are the **Trivial Baseline (training mean)**, **Simple Baseline (mileage-only OLS)**, and **Strong Baseline (200-tree random forest on Set A, seed 42; no depth limit, default leaf size 1)**. The forest is an inherited fixed flexible comparator, not a tuned optimum. All methods use identical rows and metrics. Paired test-loss gaps use 10,000 dealer-cluster bootstrap resamples (seed 42), keeping both methods' predictions together. Grouped-fold paired t-tests have only five overlapping training fits, so their p-values are exploratory.

Method	Group MAE	Test MAE	RMSE	R^2
Mean	8,552 ± 1,354	8,906	14,040	-0.004
Mileage	7,806 ± 1,171	8,116	12,909	0.152
RF (raw)	**2,660 ± 236**	**3,227**	**5,103**	**0.867**
OLS A	3,991 ± 429	4,290	5,971	0.818
OLS C (ours)	3,324 ± 267	3,711	5,399	0.852

Table 2: Main results: errors in EUR; Group MAE is five-fold mean ± sample SD. RMSE and R^2 use all 752 test rows. Bold marks the best values.

7 Results

7.1 Main comparison

Table 2 shows that engineered OLS improves over raw OLS but remains behind the random forest. Test MAE falls by EUR 578 (13.5%) from A to C; the forest's advantage over C is EUR 484. A piecewise tree ensemble can capture local trim/usage combinations beyond the three slopes and six flags, although the comparison does not isolate that mechanism. The forest remains a baseline; our final predictor is linear.

Against the proposed threshold, C exceeds EUR 3,000 by EUR 711, and RF exceeds it by EUR 227. Only 55.6% of C's test errors lie within EUR 3,000 (RF: 59.4%). The average threshold is not satisfied even though many individual estimates fall inside it. Thus these predictions are inspection aids, not autonomous pricing decisions.

7.2 Ablations and uncertainty

Set	KFold MAE	Group MAE	Test MAE
A: raw	3,939 ± 105	3,991 ± 429	4,290
B: top-5	3,966 ± 118	4,022 ± 446	4,329
C: full	3,244 ± 92	3,324 ± 267	3,711
C − flags	3,457 ± 95	3,522 ± 278	3,871
C − interactions	3,705 ± 58	3,779 ± 414	4,081

Table 3: Matched OLS ablations, EUR. All folds and preprocessing rules are shared. Lower is better; C is best in every column.

Only the feature representation changes in Table 3; every row uses OLS and the same splits. Removing either engineered group worsens dealer-grouped CV, supporting H1 and H2. Set C also improves on top-k Set B, supporting H3: ranking raw associations cannot create title information or model-specific slopes. Both groups use already collected fields, adding nine numeric terms rather than a new data source. Their costs are parsing maintenance and harder interpretation of overlapping trim mentions.

Comparison	Gap	95% cluster CI	p
A − C	578	[374, 796]	0.001
B − C	617	[407, 844]	0.002
RF − C	-484	[-752, -205]	0.001

Table 4: Test-MAE gaps (EUR): positive favors C. CI: paired dealer-cluster bootstrap, 10,000 draws; p: exploratory paired grouped-fold t-test.

The paired bootstrap intervals in Table 4 preserve within-dealer dependence in test losses. They describe uncertainty for these fixed fitted models and observed sellers; they omit retraining, feature selection and market change. Reused test data and uncertain vehicle identity prevent treating these intervals as definitive prospective evidence.

Model	n	A	B	C	Med.%	≤3k%
3	347	3,467	3,481	3,004	7.9	63
S	69	6,790	6,917	6,289	16.7	29
Y	271	3,723	3,770	3,161	6.7	58
X	59	7,816	7,876	6,423	9.5	42
Roadster	5	12,403	12,395	11,976	13.0	0
Cybertruck	1	22,051	21,858	19,203	12.0	0

Table 5: Test MAE by model (EUR). Last columns: C median absolute percentage error and share within EUR 3,000. Rare classes are not interpreted.

C improves every common-model average (Table 5), least for S. Relative to price, X (9.5%) is close to 3/Y; S is weakest (16.7%), with many old, cheap cars. Rare classes are not interpreted. No model meets the EUR 3,000 threshold; Model 3's EUR 3,004 exceeds it by EUR 4. Excluding six rare cars gives test MAE EUR 4,211 (A), 3,635 (C), and 3,090 (RF).

Sensitivity to data quality. A model/registration-month/mileage duplicate proxy flags 152 test rows overlapping training. On the 600 remaining rows, MAE is EUR 4,550 (A), EUR 4,031 (C), and EUR 3,816 (RF). Matching attributes do not prove identical vehicles; exclusion is a sensitivity analysis, not a replacement split. Removing the single Model Y/2001 training record and refitting C gives test MAE EUR 3,713. This post-hoc audit is not used to optimize the headline model.

8 Analysis

8.1 Coefficients and significance

Table 6 lists all 28 non-intercept terms. HC3 is undefined (one leverage-1 row, the single Cybertruck), so we use HC1 robust t inference. Train-

Term	β	95% CI	p	VIF
Year	4.56	[4.05, 5.07]	$< .001$	5.9
Mileage	-3.44	[-3.87, -3.02]	$< .001$	1.7
Power	2.78	[2.31, 3.24]	$< .001$	3.4
Equipment	0.19	[-0.03, 0.41]	0.090	1.3
Performance	2.94	[2.17, 3.71]	$< .001$	2.1
Long range	1.09	[0.39, 1.79]	0.002	2.0
AWD	-1.46	[-2.23, -0.69]	$< .001$	2.4
Entry trim	-1.39	[-2.08, -0.70]	$< .001$	2.9
New gen.	4.35	[3.24, 5.46]	$< .001$	1.4
HW4	5.20	[3.35, 7.06]	$< .001$	1.1
Model S×year	3.17	[2.62, 3.72]	$< .001$	4.2
Model X×year	2.96	[2.36, 3.57]	$< .001$	2.4
Model Y×year	-0.27	[-1.13, 0.59]	0.535	2.7
Model S	20.31	[18.83, 21.79]	$< .001$	3.5
Model X	24.76	[22.84, 26.68]	$< .001$	2.5
Model Y	3.24	[1.90, 4.59]	$< .001$	2.3
Roadster	85.02	[67.30, 102.75]	$< .001$	1.5
Cybertruck	133.40	—	—	1.0
Unknown	-11.69	[-20.40, -2.98]	0.009	1.0
IT	2.78	[2.09, 3.46]	$< .001$	1.5
ES	-1.43	[-2.22, -0.64]	$< .001$	1.3
BE	-0.84	[-1.56, -0.12]	0.022	1.3
DE	-0.39	[-1.01, 0.23]	0.215	1.7
FR	-0.29	[-1.15, 0.58]	0.514	1.2
AT	-0.09	[-0.76, 0.58]	0.787	1.3
LU	-0.10	[-2.76, 2.56]	0.942	1.0
FSD	0.87	[0.14, 1.59]	0.020	1.1
EAP	0.60	[-0.32, 1.52]	0.201	1.0

Table 6: Final Set C: HC1 intervals, β/CI in 1,000 EUR. Continuous/interaction terms: per SD; flags: 0-to-1; categories: reference contrasts. All 28 terms shown. Rare classes (Roadster, Cybertruck, Unknown; $n < 20$) are not interpreted; singleton uncertainty is suppressed.

ing R^2 is 0.850; max VIF 5.9 (year), moderate. HC1 ignores dealer dependence and no multiple-testing correction is applied, so results are exploratory associations. Continuous and interaction terms are per training SD; flags are 0-to-1 effects. Category references are Model 3, Netherlands and AP; the singleton's uncertainty is suppressed.

Interpretation. Each newer year adds EUR 1,882 for Model 3, about EUR 4,178 for S and EUR 4,994 for X; Y does not differ from Model 3. This supports heterogeneous slopes, not a causal effect of aging. Trim matters beyond power and model: HW4, new generation, Performance and Long Range carry premiums and the entry trim a discount (Table 6). Italy (+EUR 2,775), Spain (−EUR 1,430) and Belgium (−EUR 842) differ from the Netherlands; other countries do not. FSD adds EUR 866; EAP is not significant.

AWD (−EUR 1,461) corrects for overlap: 88% of AWD-flagged listings also carry Long Range or Performance and report more power (318 vs 282 kW), so those terms take most of the premium; a Performance AWD car still nets +EUR 1,480. Raw AWD listings are also cheaper (EUR 34,596 vs EUR 36,016), so the sign is not a reversal.

ID	Test	Δ K/G	Verdict
H1	Remove flags	213 / 198	Supported
H2	Remove interactions	462 / 455	Supported
H3	B minus C	723 / 698	Supported

Table 7: Section 4 hypotheses resolved using Table 3. Δ K/G: KFold/grouped MAE minus C, EUR. Positive favors C. B−C test gap is EUR 617 (Table 4).

Resolving the hypotheses. H1–H3 retain their Section 4 definitions. Removing flags raises KFold/grouped MAE by EUR 213/198; removing interactions raises them by EUR 462/455 (Table 3). C beats B by EUR 723/698 in those validations and EUR 617 on test. These results support H1, H2 and H3 for this benchmark, not prospective or causal claims. A group can help even if an individual coefficient is insignificant.

ID	Additional check	Δ MAE	p
E1	Add colour	32	0.320
E2	Remove country	78	0.010
E3	Remove model	2,149	$< .001$
E4	Add power squared	-107	$< .001$

Table 8: Additional exploratory checks E1–E4: KFold MAE change from C (EUR 3,244), positive = worse; paired fold t-tests. E3 removes model and its year terms. These do not redefine H1–H3.

Additional checks. E1–E4 add or remove one group and compare KFold MAE with C (paired fold t-test). Colour gives no prediction gain, although a joint Wald test finds a small in-sample link. Country matters, but little (+EUR 78, $p = 0.010$). Removing car model, with its year terms, costs EUR 2,149, by far the largest change. Price is curved in power (−EUR 107, $p < .001$); we report this without changing the frozen model. A small p-value does not make an effect useful: compare its natural-unit size to the EUR 3,000 screening tolerance.

8.2 Error analysis

Cases 1 and 4: Model X 75D, France, 2018, and Model S 70, Spain, 2016. We find comparable training maxima EUR 39,999 (27 X "75" cars) and EUR 34,000 (nine S 70 cars), below their asking prices. Unusual pricing is plausible, not verified mispricing. **Case 2:** Model X 75 Dual Motor,

Case	Actual	Predicted	Error
1	75,900	34,842	41,058
2	90,000	61,820	28,180
3	12,000	-15,506	27,506
4	48,600	24,643	23,957
5	32,990	9,683	23,307

Table 9: Five largest absolute errors of Set C (EUR). Error = actual − predicted. Listing context below.

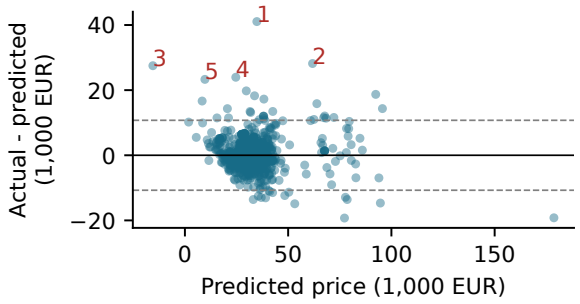


Figure 3: Set C test residuals, actual minus predicted EUR. Labels mark the five largest absolute errors in Table 9; all 752 test rows are shown; dashed lines mark ±2 residual SD.

Belgium, 2023: the title and 244 kW suggest an older 75D, conflicting with registration. **Case 3:** Model S 85, Netherlands, 2014: 670,000 km exceeds the training maximum 635,000; linear extrapolation produces the only negative prediction. **Case 5:** Model S P85 RWD, Spain, 2013: price is near the comparable median EUR 26,170, but the Performance pattern misses "P85", "RWD" activates entry trim, and the S slope lowers the prediction.

Shared pattern. Old S cars are weakest (Cases 3–5; 16.7% median percentage error). The constant euro/year slope may be too steep at old ages. We find 70% of the top 20 errors are underpredictions (residual skew 1.14). Proposed checks include flattening age effects, a minimum-price clip, P85/P90 parsing and title/power/year consistency. These test-inspired proposals are not applied, to avoid further test tuning.

9 Conclusion and Limitations

Trim flags and model-specific year slopes reduce test MAE to EUR 3,711, versus EUR 4,290 for raw OLS and EUR 3,227 for RF. H1–H3 are supported within this benchmark, but neither C nor RF meets the assumed EUR 3,000 tolerance. Use estimates to prioritize inspections, verifying trim, battery condition and specifications.

One website, eight countries and one asking-price snapshot limit coverage. The supplied order split, prior test exposure, possible duplicates, inherited annotations and rare classes limit generalization. EUR 3,000 lacks stakeholder cost evidence and is stricter for cheaper cars. Linear extrapolation can be implausible; associations are not causal depreciation. Next, freeze the rules and collect a locked snapshot with vehicle-identity/battery checks, evaluating accuracy and actual screening decisions.

Contributions of Each Member

Group A: 馮昱祥 (515559007), 林暉庭 (314551160) jointly prepared the abstract/introduction and research question, reviewed related work, collected/preprocessed data, ran EDA, and designed the feature pipeline and figure.
Group B: 杜洺鋒 (315554001), 邱育宏 (315554053), 吳宥駒 (315551146) jointly handled model choice/tuning, validation, metrics/threshold/baselines, and main comparisons with significance testing.
Group C: 黃少鋒 (315561001), 莊誌強 (315561002) jointly ran ablations, interpreted coefficients/significance/VIF, analysed the largest errors, and wrote conclusions/limitations.

AI usage. ChatGPT/Codex assisted with code, recomputation, figures, literature checking, writing and compilation. The team supplied data/feature definitions and remains responsible for verification.

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